Prerequisites

Before attending this course you should have attended and passed the department's introductory quantum physics courses, PHAS0022, or equivalent. Frequent reference will be made to the material in PHAS0022. The following topics should have been covered previously:

Quantum Mechanics

- The **time-independent Schrödinger wave equation** and its solution for:
  - quantum **wells** and quantum **barriers**/steps
  - The harmonic **oscillator** (classical and quantum)
  - The **hydrogen atom** including the radial equation as well as the angular equation and its solution with spherical harmonics.

  An understanding of atomic spectroscopic notation (n, l, m quantum numbers) and their physical basis is assumed.
- The **expansion postulate**; the **Born interpretation** of the wave function; simple calculations of **probabilities** and **expectation values**.
- For a time-independent Hamiltonian, an understanding of the separability of the full Schrödinger equation into a **time-independent wave equation** in position space and a **time-dependent component** is assumed. Familiarity with applications to eigenstates and/or superpositions thereof is assumed.
- The **postulates** of quantum theory.
- The **mathematical techniques** used to describe the above (complex numbers, linear differential equations).

Here is a list of the vector and matrix techniques (linear algebra) that we will be using in this course. You may have learnt these in the first year so they may be rusty. If so please revise them!

Vectors

- What is a **vector** and how we write one down.
- How we **multiply** vectors with **scalars** and **add vectors together**.
- What is a **vector space**.
- What is a **normalised vector** and how do we **normalise** a vector.

Matrices

- What is a **matrix** and how we write one down.
- How we **multiply matrices with scalars** and **add matrices together**
- How we **multiply matrices** together.

Manipulating matrices

- Taking the **complex conjugate** of a matrix
- Taking the **transpose** of a matrix or a product of matrices
- Taking the **Hermitian conjugate** of a matrix or a product of matrices
- Calculating the **commutator** of 2 matrices and knowing when matrices commute

Types of matrices

- Knowing the definition of:
  - **Diagonal** matrices
  - **Hermitian** matrices
  - **Unitary** matrices

Eigenvalues and eigenvectors

- The definition of the **eigenvalues** and **eigenvectors** of a matrix
- Calculating the eigenvalues of a matrix by solving the **characteristic equation**.
- Calculating **normalised** eigenvectors
- **Diagonal** matrices - how to write down eigenvalues and eigenvectors without any calculation
- The properties of eigenvalues and eigenvectors of Hermitian matrices

If you want to revise the quantum physics topics we recommend the PHAS0022 lecture notes or:
A.I.M. Rae, Quantum Mechanics, Fifth edition, (Taylor & Francis 2007).
For the mathematics topics, again, your previous mathematics courses will be helpful, or chapter 3 of:
Mary Boas, Mathematical Methods in the Physical Sciences, John Wiley & Sons; 3rd Edition (26 Aug. 2005)